

Newman-Penrose Formalism Calculations

Stephani et al., Equation (33.38)

Metric

$$ds^2 = -dt \otimes dt + e^{(2b)} dx \otimes dx + e^{(2b)} dy \otimes dy + e^{(2a)} dz \otimes dz$$

Coordinates

$$(x^\mu) = (t, x, y, z)$$

Null Tetrad

Covariant components

$$l = \left(\frac{1}{2} \sqrt{2} \right) dt + \left(\frac{1}{2} \sqrt{2} e^b \right) dx$$

$$n = \left(\frac{1}{2} \sqrt{2} \right) dt + \left(-\frac{1}{2} \sqrt{2} e^b \right) dx$$

$$m = \left(\frac{1}{2} \sqrt{2} e^b \right) dy + \left(\frac{1}{2} i \sqrt{2} e^a \right) dz$$

$$\bar{m} = \left(\frac{1}{2} \sqrt{2} e^b \right) dy + \left(-\frac{1}{2} i \sqrt{2} e^a \right) dz$$

Contravariant components

$$l = \left(-\frac{1}{2} \sqrt{2} \right) dt + \left(\frac{1}{2} \sqrt{2} e^{(-b)} \right) dx$$

$$n = \left(-\frac{1}{2}\sqrt{2}\right) dt + \left(-\frac{1}{2}\sqrt{2}e^{(-b)}\right) dx$$

$$m = \left(\frac{1}{2}\sqrt{2}e^{(-b)}\right) dy + \left(\frac{1}{2}i\sqrt{2}e^{(-a)}\right) dz$$

$$\overline{m} = \left(\frac{1}{2}\sqrt{2}e^{(-b)}\right) dy + \left(-\frac{1}{2}i\sqrt{2}e^{(-a)}\right) dz$$

Directional Derivatives

$$D = -\frac{1}{2}\sqrt{2}\partial_t + \frac{1}{2}\sqrt{2}e^{(-b)}\partial_x$$

$$\Delta = -\frac{1}{2}\sqrt{2}\partial_t - \frac{1}{2}\sqrt{2}e^{(-b)}\partial_x$$

$$\delta = \frac{1}{2}\sqrt{2}e^{(-b)}\partial_y + \frac{1}{2}i\sqrt{2}e^{(-a)}\partial_z$$

$$\bar{\delta} = \frac{1}{2}\sqrt{2}e^{(-b)}\partial_y - \frac{1}{2}i\sqrt{2}e^{(-a)}\partial_z$$

Spin Coefficients

$$\rho = -\frac{1}{4}\sqrt{2}e^{(-b)}\frac{\partial}{\partial x}a - \frac{1}{4}\sqrt{2}e^{(-b)}\frac{\partial}{\partial x}b$$

$$\sigma = \frac{1}{4}\sqrt{2}e^{(-b)}\frac{\partial}{\partial x}a - \frac{1}{4}\sqrt{2}e^{(-b)}\frac{\partial}{\partial x}b$$

$$\kappa = \frac{1}{4}\sqrt{2}e^{(-b)}\frac{\partial}{\partial y}b + \frac{1}{4}i\sqrt{2}e^{(-a)}\frac{\partial}{\partial z}b$$

$$\mu = -\frac{1}{4} \sqrt{2} e^{(-b)} \frac{\partial}{\partial x} a - \frac{1}{4} \sqrt{2} e^{(-b)} \frac{\partial}{\partial x} b$$

$$\lambda = \frac{1}{4} \sqrt{2} e^{(-b)} \frac{\partial}{\partial x} a - \frac{1}{4} \sqrt{2} e^{(-b)} \frac{\partial}{\partial x} b$$

$$\nu = -\frac{1}{4} \sqrt{2} e^{(-b)} \frac{\partial}{\partial y} b + \frac{1}{4} i \sqrt{2} e^{(-a)} \frac{\partial}{\partial z} b$$

$$\alpha = -\frac{1}{4} \sqrt{2} e^{(-b)} \frac{\partial}{\partial y} a + \frac{1}{4} i \sqrt{2} e^{(-a)} \frac{\partial}{\partial z} b$$

$$\beta = \frac{1}{4} \sqrt{2} e^{(-b)} \frac{\partial}{\partial y} a + \frac{1}{4} i \sqrt{2} e^{(-a)} \frac{\partial}{\partial z} b$$

$$\tau = -\frac{1}{4} \sqrt{2} e^{(-b)} \frac{\partial}{\partial y} b - \frac{1}{4} i \sqrt{2} e^{(-a)} \frac{\partial}{\partial z} b$$

$$\pi = \frac{1}{4} \sqrt{2} e^{(-b)} \frac{\partial}{\partial y} b - \frac{1}{4} i \sqrt{2} e^{(-a)} \frac{\partial}{\partial z} b$$

$$\epsilon = 0$$

$$\gamma = 0$$

Ricci Tensor Components

$$\Phi_{00} = -\frac{1}{4} e^{(-2b)} \frac{\partial}{\partial x} a^2 + \frac{1}{4} e^{(-2b)} \frac{\partial}{\partial x} a \frac{\partial}{\partial x} b - \frac{1}{4} e^{(-2b)} \frac{\partial}{\partial y} a \frac{\partial}{\partial y} b + \frac{1}{4} e^{(-2a)} \frac{\partial}{\partial z} a \frac{\partial}{\partial z} b - \frac{1}{2} e^{(-2a)} \frac{\partial}{\partial z} b^2 - \frac{1}{4} e^{(-2b)} \frac{\partial^2}{(\partial x)^2} a - \frac{1}{4} e^{(-2b)} \frac{\partial^2}{(\partial x)^2} b - \frac{1}{4} e^{(-2b)} \frac{\partial^2}{(\partial y)^2} b - \frac{1}{4} e^{(-2a)} \frac{\partial^2}{(\partial z)^2} b$$

$$\Phi_{01} = -\frac{1}{4} e^{(-2b)} \frac{\partial}{\partial x} a \frac{\partial}{\partial y} a + \frac{1}{4} e^{(-2b)} \frac{\partial}{\partial y} a \frac{\partial}{\partial x} b + \frac{1}{4} e^{(-2b)} \frac{\partial}{\partial x} a \frac{\partial}{\partial y} b + \frac{1}{4} i e^{(-a-b)} \frac{\partial}{\partial x} a \frac{\partial}{\partial z} b - \frac{1}{4} e^{(-2b)} \frac{\partial^2}{\partial x \partial y} a - \frac{1}{4} i e^{(-a-b)} \frac{\partial^2}{\partial x \partial z} b$$

$$\begin{aligned}\Phi_{02} = & \frac{1}{4} e^{(-2b)} \frac{\partial}{\partial x} a^2 - \frac{1}{4} e^{(-2b)} \frac{\partial}{\partial x} a \frac{\partial}{\partial x} b + \frac{1}{4} e^{(-2b)} \frac{\partial}{\partial y} a \frac{\partial}{\partial y} b + \frac{1}{2} i e^{(-a-b)} \frac{\partial}{\partial y} a \frac{\partial}{\partial z} b - \frac{1}{4} e^{(-2a)} \frac{\partial}{\partial z} a \frac{\partial}{\partial z} b \\ & + \frac{1}{4} e^{(-2b)} \frac{\partial^2}{(\partial x)^2} a - \frac{1}{4} e^{(-2b)} \frac{\partial^2}{(\partial x)^2} b - \frac{1}{4} e^{(-2b)} \frac{\partial^2}{(\partial y)^2} b - \frac{1}{2} i e^{(-a-b)} \frac{\partial^2}{\partial y \partial z} b + \frac{1}{4} e^{(-2a)} \frac{\partial^2}{(\partial z)^2} b\end{aligned}$$

$$\Phi_{10} = -\frac{1}{4}e^{(-2b)}\frac{\partial}{\partial x}a\frac{\partial}{\partial y}a + \frac{1}{4}e^{(-2b)}\frac{\partial}{\partial y}a\frac{\partial}{\partial x}b + \frac{1}{4}e^{(-2b)}\frac{\partial}{\partial x}a\frac{\partial}{\partial y}b - \frac{1}{4}ie^{(-a-b)}\frac{\partial}{\partial x}a\frac{\partial}{\partial z}b - \frac{1}{4}e^{(-2b)}\frac{\partial^2}{\partial x\partial y}a + \frac{1}{4}ie^{(-a-b)}\frac{\partial^2}{\partial x\partial z}b$$

$$\Phi_{11} = -\frac{1}{4} e^{(-2b)} \frac{\partial}{\partial y} a^2 - \frac{1}{4} e^{(-2b)} \frac{\partial}{\partial x} a \frac{\partial}{\partial x} b + \frac{1}{4} e^{(-2b)} \frac{\partial}{\partial y} a \frac{\partial}{\partial y} b + \frac{1}{4} e^{(-2a)} \frac{\partial}{\partial z} a \frac{\partial}{\partial z} b - \frac{1}{4} e^{(-2a)} \frac{\partial}{\partial z} b^2 - \frac{1}{4} e^{(-2b)} \frac{\partial^2}{(\partial y)^2} a - \frac{1}{4} e^{(-2a)} \frac{\partial^2}{(\partial z)^2} b$$

$$\Phi_{12} = \frac{1}{4} e^{(-2b)} \frac{\partial}{\partial x} a \frac{\partial}{\partial y} a - \frac{1}{4} e^{(-2b)} \frac{\partial}{\partial y} a \frac{\partial}{\partial x} b - \frac{1}{4} e^{(-2b)} \frac{\partial}{\partial x} a \frac{\partial}{\partial y} b - \frac{1}{4} i e^{(-a-b)} \frac{\partial}{\partial x} a \frac{\partial}{\partial z} b + \frac{1}{4} e^{(-2b)} \frac{\partial^2}{\partial x \partial y} a + \frac{1}{4} i e^{(-a-b)} \frac{\partial^2}{\partial x \partial z} b$$

$$\begin{aligned}\Phi_{20} = & \frac{1}{4} e^{(-2b)} \frac{\partial}{\partial x} a^2 - \frac{1}{4} e^{(-2b)} \frac{\partial}{\partial x} a \frac{\partial}{\partial x} b + \frac{1}{4} e^{(-2b)} \frac{\partial}{\partial y} a \frac{\partial}{\partial y} b - \frac{1}{2} i e^{(-a-b)} \frac{\partial}{\partial y} a \frac{\partial}{\partial z} b - \frac{1}{4} e^{(-2a)} \frac{\partial}{\partial z} a \frac{\partial}{\partial z} b \\ & + \frac{1}{4} e^{(-2b)} \frac{\partial^2}{(\partial x)^2} a - \frac{1}{4} e^{(-2b)} \frac{\partial^2}{(\partial x)^2} b - \frac{1}{4} e^{(-2b)} \frac{\partial^2}{(\partial y)^2} b + \frac{1}{2} i e^{(-a-b)} \frac{\partial^2}{\partial y \partial z} b + \frac{1}{4} e^{(-2a)} \frac{\partial^2}{(\partial z)^2} b\end{aligned}$$

$$\Phi_{21} = \frac{1}{4} e^{(-2b)} \frac{\partial}{\partial x} a \frac{\partial}{\partial y} a - \frac{1}{4} e^{(-2b)} \frac{\partial}{\partial y} a \frac{\partial}{\partial x} b - \frac{1}{4} e^{(-2b)} \frac{\partial}{\partial x} a \frac{\partial}{\partial y} b + \frac{1}{4} i e^{(-a-b)} \frac{\partial}{\partial x} a \frac{\partial}{\partial z} b + \frac{1}{4} e^{(-2b)} \frac{\partial^2}{\partial x \partial y} a - \frac{1}{4} i e^{(-a-b)} \frac{\partial^2}{\partial x \partial z} b$$

$$\Phi_{22} = -\frac{1}{4} e^{(-2b)} \frac{\partial}{\partial x} a^2 + \frac{1}{4} e^{(-2b)} \frac{\partial}{\partial x} a \frac{\partial}{\partial x} b - \frac{1}{4} e^{(-2b)} \frac{\partial}{\partial y} a \frac{\partial}{\partial y} b + \frac{1}{4} e^{(-2a)} \frac{\partial}{\partial z} a \frac{\partial}{\partial z} b - \frac{1}{2} e^{(-2a)} \frac{\partial}{\partial z} b^2 - \frac{1}{4} e^{(-2b)} \frac{\partial^2}{(\partial x)^2} a - \frac{1}{4} e^{(-2b)} \frac{\partial^2}{(\partial x)^2} b - \frac{1}{4} e^{(-2b)} \frac{\partial^2}{(\partial y)^2} b - \frac{1}{4} e^{(-2a)} \frac{\partial^2}{(\partial z)^2} b$$

$$\Lambda = -\frac{1}{12} e^{(-2b)} \frac{\partial}{\partial x} a^2 - \frac{1}{12} e^{(-2b)} \frac{\partial}{\partial y} a^2 + \frac{1}{6} e^{(-2a)} \frac{\partial}{\partial z} a \frac{\partial}{\partial z} b - \frac{1}{4} e^{(-2a)} \frac{\partial}{\partial z} b^2 - \frac{1}{12} e^{(-2b)} \frac{\partial^2}{(\partial x)^2} a - \frac{1}{12} e^{(-2b)} \frac{\partial^2}{(\partial y)^2} a - \frac{1}{12} e^{(-2b)} \frac{\partial^2}{(\partial x)^2} b - \frac{1}{12} e^{(-2b)} \frac{\partial^2}{(\partial y)^2} b - \frac{1}{6} e^{(-2a)} \frac{\partial^2}{(\partial z)^2} b$$

Weyl Tensor Components

$$\begin{aligned}\Psi_0 = & \frac{1}{4} e^{(-2b)} \frac{\partial}{\partial x} a^2 - \frac{1}{4} e^{(-2b)} \frac{\partial}{\partial x} a \frac{\partial}{\partial x} b + \frac{1}{4} e^{(-2b)} \frac{\partial}{\partial y} a \frac{\partial}{\partial y} b + \frac{1}{2} i e^{(-a-b)} \frac{\partial}{\partial y} a \frac{\partial}{\partial z} b - \frac{1}{4} e^{(-2a)} \frac{\partial}{\partial z} a \frac{\partial}{\partial z} b \\ & + \frac{1}{4} e^{(-2b)} \frac{\partial^2}{(\partial x)^2} a - \frac{1}{4} e^{(-2b)} \frac{\partial^2}{(\partial x)^2} b - \frac{1}{4} e^{(-2b)} \frac{\partial^2}{(\partial y)^2} b - \frac{1}{2} i e^{(-a-b)} \frac{\partial^2}{\partial y \partial z} b + \frac{1}{4} e^{(-2a)} \frac{\partial^2}{(\partial z)^2} b\end{aligned}$$

$$\Psi_1 = \frac{1}{4} e^{(-2b)} \frac{\partial}{\partial x} a \frac{\partial}{\partial y} a - \frac{1}{4} e^{(-2b)} \frac{\partial}{\partial y} a \frac{\partial}{\partial x} b - \frac{1}{4} e^{(-2b)} \frac{\partial}{\partial x} a \frac{\partial}{\partial y} b - \frac{1}{4} i e^{(-a-b)} \frac{\partial}{\partial x} a \frac{\partial}{\partial z} b + \frac{1}{4} e^{(-2b)} \frac{\partial^2}{\partial x \partial y} a + \frac{1}{4} i e^{(-a-b)} \frac{\partial^2}{\partial x \partial z} b$$

$$\begin{aligned}\Psi_2 = & -\frac{1}{12} e^{(-2b)} \frac{\partial}{\partial x} a^2 + \frac{1}{6} e^{(-2b)} \frac{\partial}{\partial y} a^2 + \frac{1}{4} e^{(-2b)} \frac{\partial}{\partial x} a \frac{\partial}{\partial x} b - \frac{1}{4} e^{(-2b)} \frac{\partial}{\partial y} a \frac{\partial}{\partial y} b - \frac{1}{12} e^{(-2a)} \frac{\partial}{\partial z} a \frac{\partial}{\partial z} b \\ & - \frac{1}{12} e^{(-2b)} \frac{\partial^2}{(\partial x)^2} a + \frac{1}{6} e^{(-2b)} \frac{\partial^2}{(\partial y)^2} a - \frac{1}{12} e^{(-2b)} \frac{\partial^2}{(\partial x)^2} b - \frac{1}{12} e^{(-2b)} \frac{\partial^2}{(\partial y)^2} b + \frac{1}{12} e^{(-2a)} \frac{\partial^2}{(\partial z)^2} b\end{aligned}$$

$$\Psi_3 = -\frac{1}{4} e^{(-2b)} \frac{\partial}{\partial x} a \frac{\partial}{\partial y} a + \frac{1}{4} e^{(-2b)} \frac{\partial}{\partial y} a \frac{\partial}{\partial x} b + \frac{1}{4} e^{(-2b)} \frac{\partial}{\partial x} a \frac{\partial}{\partial y} b - \frac{1}{4} i e^{(-a-b)} \frac{\partial}{\partial x} a \frac{\partial}{\partial z} b - \frac{1}{4} e^{(-2b)} \frac{\partial^2}{\partial x \partial y} a + \frac{1}{4} i e^{(-a-b)} \frac{\partial^2}{\partial x \partial z} b$$

$$\begin{aligned}\Psi_4 = & \frac{1}{4} e^{(-2b)} \frac{\partial}{\partial x} a^2 - \frac{1}{4} e^{(-2b)} \frac{\partial}{\partial x} a \frac{\partial}{\partial x} b + \frac{1}{4} e^{(-2b)} \frac{\partial}{\partial y} a \frac{\partial}{\partial y} b - \frac{1}{2} i e^{(-a-b)} \frac{\partial}{\partial y} a \frac{\partial}{\partial z} b - \frac{1}{4} e^{(-2a)} \frac{\partial}{\partial z} a \frac{\partial}{\partial z} b \\ & + \frac{1}{4} e^{(-2b)} \frac{\partial^2}{(\partial x)^2} a - \frac{1}{4} e^{(-2b)} \frac{\partial^2}{(\partial x)^2} b - \frac{1}{4} e^{(-2b)} \frac{\partial^2}{(\partial y)^2} b + \frac{1}{2} i e^{(-a-b)} \frac{\partial^2}{\partial y \partial z} b + \frac{1}{4} e^{(-2a)} \frac{\partial^2}{(\partial z)^2} b\end{aligned}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "I"